Complex numbers: the algebraic point of view

Lecture and worked examples

Objectives

Compute with the algebraic form; use the conjugate to obtain an inverse or a quotient; solve a linear equation, or a simple equation involving z and $\bar{z}$; expand a power with the binomial formula.

Introductory activity

In the sixteenth century, a formula was known for solving equations of the form $x^3 = px + q$. It is given here and taken for granted:

$$x = \sqrt[3]{\frac{q}{2} + \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}} + \sqrt[3]{\frac{q}{2} - \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}}.$$

We study the equation $x^3 = 15x + 4$.

Introductory activity (2)

1. Check that 4 is a solution of the equation.
2. Compute $\frac{q^2}{4} - \frac{p^3}{27}$ for $p = 15$ and $q = 4$. What becomes of the formula? Can it be applied in $\mathbb{R}$?
3. A calculator of the time decides to carry on as if a number written $\sqrt{-1}$ existed, with the single rule $\sqrt{-1} \times \sqrt{-1} = -1$ and the usual rules of computation for everything else. Expand $\left(2 + \sqrt{-1}\right)^3$ with these rules.

Introductory activity (3)

4. Assume that $\sqrt{-121}$ can be written $11\sqrt{-1}$. Show that the first term of the formula can be taken equal to $2 + \sqrt{-1}$, then find in the same way a value for the second term.
5. What does the formula then give? Compare with question 1.
6. Is this computation a proof? Which rules would have to be fixed for it to become one?

The set of complex numbers

We take for granted the existence of a set written $\mathbb{C}$, called the set of *complex numbers*, such that:

1. $\mathbb{C}$ contains $\mathbb{R}$;
2. $\mathbb{C}$ contains a number written i such that $i^2 = -1$;
3. $\mathbb{C}$ has an addition and a multiplication that extend those of $\mathbb{R}$ and follow the same rules of computation (commutativity, associativity, distributivity);
4. every complex number z can be written in a unique way as $z = a + ib$ with a and b real.

The set of complex numbers (2)

The expression $z = a + ib$ is the *algebraic form* of z. The real number a is the *real part* of z, written $\mathrm{Re}(z)$; the real number b is the *imaginary part* of z, written $\mathrm{Im}(z)$. If $b = 0$, z is real; if $a = 0$, z is *purely imaginary*.

The set of complex numbers

Two complex numbers are equal if and only if they have the same real part and the same imaginary part. In particular, $a + ib = 0$ with a, b real is equivalent to $a = b = 0$.

The set of complex numbers

The imaginary part of $3 - 5i$ is -5, not $-5i$: it is a real number. There is no order compatible with the operations on $\mathbb{C}$: we never write $z < z'$ for non-real complex numbers.

The set of complex numbers

$(2+3i)+(1-i)=3+2i$ and $(2+3i)(1-i)=2-2i+3i-3i^2=5+i$.

The powers of i repeat every four: $i^2=-1$, $i^3=-i$, $i^4=1$. Hence $i^{4k+r}=i^r$ for every integer $k\geq 0$.

Conjugate

The *conjugate* of $z = a + ib$ (a, b real) is $\bar{z} = a - ib$.

Conjugate

For every $z = a + ib$:

$$z + \bar{z} = 2\,\mathrm{Re}(z), \qquad z - \bar{z} = 2i\,\mathrm{Im}(z), \qquad z\bar{z} = a^2 + b^2.$$

Hence z is real if and only if $\bar{z} = z$, and purely imaginary if and only if $\bar{z} = -z$. The product $z\bar{z}$ is a nonnegative real number, zero only for $z = 0$.

Conjugate

$z\bar{z} = (a + ib)(a - ib) = a^2 - (ib)^2 = a^2 + b^2$. The first two equalities are obtained by adding or subtracting $a + ib$ and $a - ib$.

Conjugate

Every nonzero complex number $z = a + ib$ has an inverse:

$$\frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{a - ib}{a^2 + b^2}.$$

Conjugate

Multiply the numerator and the denominator by the conjugate of the denominator. Example: $\frac{1}{3+4i} = \frac{3-4i}{9+16} = \frac{3}{25} - \frac{4}{25}i.$

Conjugate

For all complex numbers z, z' and every natural number n:

$$\overline{z+z'} = \overline{z} + \overline{z'}, \qquad \overline{zz'} = \overline{z}\,\overline{z'}, \qquad \overline{z^n} = \overline{z}^n,$$

and if $z' \neq 0$: $\overline{\left(\frac{1}{z'}\right)} = \frac{1}{\overline{z'}}$ and $\overline{\left(\frac{z}{z'}\right)} = \frac{\overline{z}}{\overline{z'}}$.

Conjugate

Product. Let $z = a + ib$ and $z' = a' + ib'$. Then $zz' = (aa' - bb') + i(ab' + a'b)$, so $\overline{zz'} = (aa' - bb') - i(ab' + a'b)$. On the other hand, $\overline{z}\,\overline{z'} = (a - ib)(a' - ib') = (aa' - bb') - i(ab' + a'b)$. The two results are equal.

Inverse. If $z' \neq 0$, $z' \times \dfrac{1}{z'} = 1$. By the property of the product, $\overline{z'} \times \overline{\left(\dfrac{1}{z'}\right)} = \overline{1} = 1$, so $\overline{\left(\dfrac{1}{z'}\right)}$ is the inverse of $\overline{z'}$. The quotient follows by writing $\dfrac{z}{z'} = z \times \dfrac{1}{z'}$.

Conjugate (2)

Power. By induction on n. For $n = 0$: $\overline{z^0} = \overline{1} = 1 = \overline{z}^0$. If $\overline{z^n} = \overline{z}^n$ for some integer n, then $\overline{z^{n+1}} = \overline{z^n \times z} = \overline{z^n} \times \overline{z} = \overline{z}^n \times \overline{z} = \overline{z}^{n+1}$.

Equations

If $a \neq 0$, the equation $az = b$ has a unique solution $z = \dfrac{b}{a}$, which we write in algebraic form.

Example: $(1 + i)z = 3 - i$ gives $z = \dfrac{(3 - i)(1 - i)}{2} = \dfrac{2 - 4i}{2} = 1 - 2i$.

Equations

Let $z = x + iy$ with x, y real, write each side in algebraic form, then identify real parts and imaginary parts. This gives a system of two real equations.

Equations

With $z = x + iy$: $2x + 2iy + i(x - iy) = (2x + y) + i(x + 2y)$. Identification gives $2x + y = 3$ and $x + 2y = -3$, hence $x = 3$ and $y = -3$. The solution is $z = 3 - 3i$. Check:
$2(3 - 3i) + i(3 + 3i) = 6 - 6i + 3i - 3 = 3 - 3i$.

Binomial formula

For all complex numbers a, b and every natural number n:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

Binomial formula

By induction on n. For $n = 0$, both sides equal 1. Assume the formula holds for some integer n. Then

$$(a+b)^{n+1} = (a+b)\sum_{k=0}^{n}\binom{n}{k}a^k b^{n-k} = \sum_{k=0}^{n}\binom{n}{k}a^{k+1}b^{n-k} + \sum_{k=0}^{n}\binom{n}{k}a^k b^{n+1-k}.$$

In the first sum, set $j = k+1$: it becomes $\sum_{j=1}^{n+1}\binom{n}{j-1}a^j b^{n+1-j}$. Grouping the terms with the same exponent of a and using Pascal's rule $\binom{n}{j-1} + \binom{n}{j} = \binom{n+1}{j}$ for $1 \le j \le n$, we obtain

$$(a+b)^{n+1} = b^{n+1} + \sum_{j=1}^{n}\binom{n+1}{j}a^j b^{n+1-j} + a^{n+1} = \sum_{j=0}^{n+1}\binom{n+1}{j}a^j b^{n+1-j}.$$

The formula holds for $n+1$, hence for every natural number n.

Binomial formula

$(1+i)^4 = 1 + 4i + 6i^2 + 4i^3 + i^4 = 1 + 4i - 6 - 4i + 1 = -4$. One can also write $(1+i)^2 = 2i$, then $(1+i)^4 = (2i)^2 = -4$.

Binomial formula

Unique algebraic form; $i^2 = -1$; $z\bar{z} = a^2 + b^2$; for a quotient, multiply by the conjugate of the denominator; for an equation in z and $\bar{z}$, set $z = x + iy$ and identify.